

USB 2.0 high-speed data lines require a nominal differential impedance of 90 ohms ($\pm 15\%$)

The diagram illustrates a USB 2.0 high-speed connector assembly. A 10-pin connector, labeled J3, is shown with pins numbered 1 through 10. The pins are connected as follows:

- Pins 1 and 2 are connected to a PWR_FLAG signal, which is also connected to GND.
- Pins 3 and 4 are connected to GND.
- Pins 5 and 6 form a differential pair labeled USB1_DP and USB1_DN.
- Pins 7 and 8 form a differential pair labeled USB2_DP and USB2_DN.
- Pins 9 and 10 are connected to GND.

The assembly is powered by +5V lines connected to the USB1 and USB2 differential pairs. The part number ESQ-105-23-T-D is indicated at the bottom.

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The schematic diagram illustrates the internal wiring of the USB-TTL-3V0 module. At the center is the FT232RL integrated circuit. The USB interface is connected to pins 1 through 5 of the CN5 connector: USB1_DP (pin 1), USB1_DN (pin 2), GND (pin 3), USB0+ (pin 4), and USB0- (pin 5). The FTDI interface is connected to pins 1 through 5 of the J2 connector: TX (pin 1), RX (pin 2), GND (pin 3), TX (pin 4), and RX (pin 5). The module is powered by a 3V3OUT supply (pin 17) and a RESET signal (pin 19). Various capacitors (C1, C2, C3, C4) are used for decoupling and timing. The diagram also shows the connection of the module to a USB cable and an FTDI cable, with a note indicating that the TX and RX pins of the FTDI cable should be connected to the TX and RX pins of the module, respectively.

Reference designs

Please note that in the schematic pin 1 is set floating as the document of the ftdi cable said not connect. Please confirm????

Page 9 CN5 table: Serial Port Connector

1---+5V---+5V
2---TX---Serial TX
3---RX---Serial RX
4---GND---Earth wire

It is assumed that TX in the table is the TX of the board and RX from FTDI needs to be connected. Please confirm. It is assumed that RX in the table is the RX of hte board and TX from FTDI needs to be connected. Please confirm.

In the wiring list given in USB-TTL-3V0 -20241203 pin 1 is not connected. To be check and proceed accordingly.

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USB OTG (On-The-Go) data lines require a 90-ohm (Ω) differential characteristic impedance with a standard tolerance of $\pm 10\%$ to $\pm 15\%$. The USB OTG has a different power net not connected to the 5V from the other 2 USBs. Please confirm.

The left diagram shows a component labeled TSM-102-01-L-DH-A. It has four pins: 01 (green) connected to PWR_FLAG, 02 (red) connected to USB_OTG_DP, 03 (green) connected to GND, and 04 (red) connected to USB_OTG_DN. A J4 connector is shown at the top with pins 01, 02, 03, and 04.

The right diagram shows a component labeled USB-A-S-X-SM2-R. It has four pins: 01 (red) connected to VBUS, 02 (red) connected to VCC, 03 (green) connected to D+, and 04 (red) connected to GND. A J5 connector is shown at the top with pins 01, 02, 03, and 04. The component also has two shield pins, S1 and S2, connected to a common shield point.